Cloud Computing for ABC Enterprise

Key Contributions to Business Automation

CLOUD COMPUTING FUNDAMENTALS (CCF501)
ASSESSMENT 1

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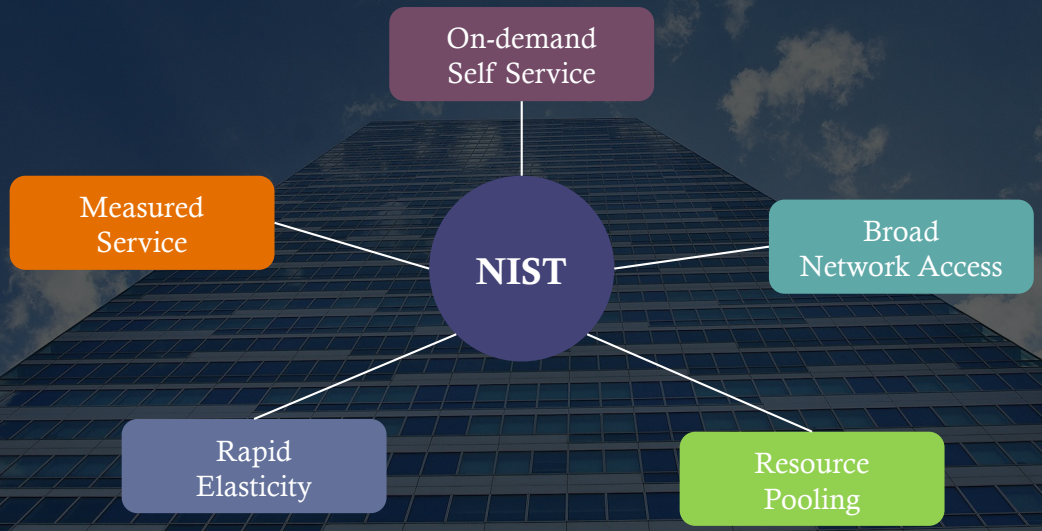
WHAT IS CLOUD COMPUTING?

On-demand delivery of compute, storage and services over the internet (Mell & Grance, 2011).



Key shift: from **owning capacity → renting capability**

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Benefits of Cloud Computing

B1 – Cost Efficiency

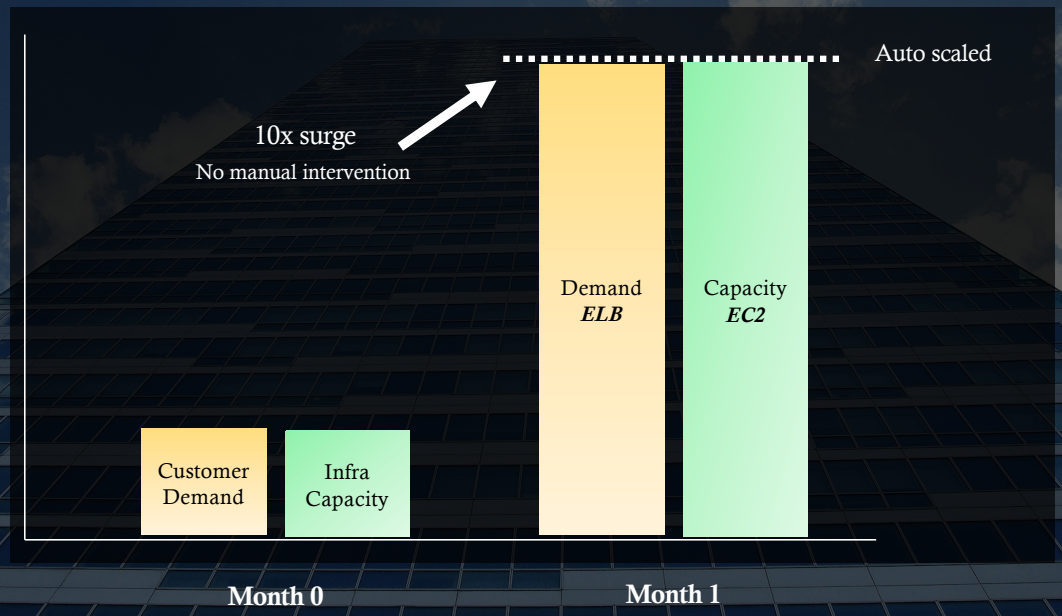
Traditional IT (CapEx)	Cloud – ABC (OpEx)
Large upfront hardware purchase	**No upfront servers**
Idle capacity during low demand	**Pay per compute hour + GB-month**
Full ops team required	**Provider manages infrastructure**
Fixed costs regardless of usage	**~80% reduction in start-up IT costs**

COST SHIFT

BENEFITS OF CLOUD COMPUTING

B2 & B3 – Scale without Limit & Less Overhead

Rapid Elasticity: ABC's 10x Growth absorbed by Auto Scaling



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Challenges of Cloud Migration

Challenge	*Example*	*Mitigation*
Security & Privacy	PII + payment data	IAM least-privilege, MFA, encryption, shared responsibility model
Cost Volatility	Pay-as-you-go can spiral without guardrails	FinOps: budget alerts, resource tagging, rightsizing, reserved pricing
Vendor Lock-in + Skills Gap	Migration cost + mindset shift	Portability-first (containers, standard DBs), targeted cloud upskilling

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Recommended Models for ABC

Cloud Service Models

IaaS
Compute + Config
Flexibility

PaaS
Managed DB, load balancing,
serverless

SaaS
Limited customization

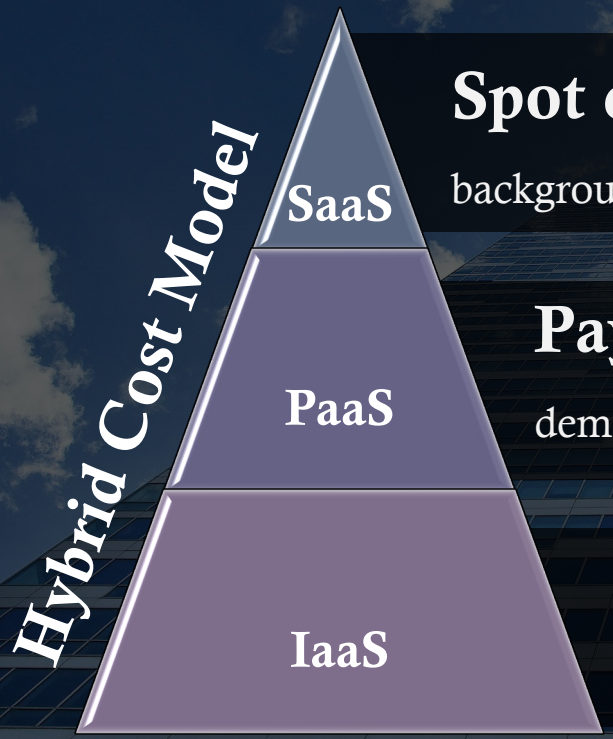
Cloud Deployment Models

Private cloud
High cost, limited elasticity

Hybrid cloud
Premature complexity

Public cloud
Scalable, automated,
cost-effective

Recommended Cost Model



Spot capacity

background jobs + analytics pipelines

Pay-as-you-go

demand spikes via Auto Scaling

Reserved

stable tiers (web/app servers, databases)

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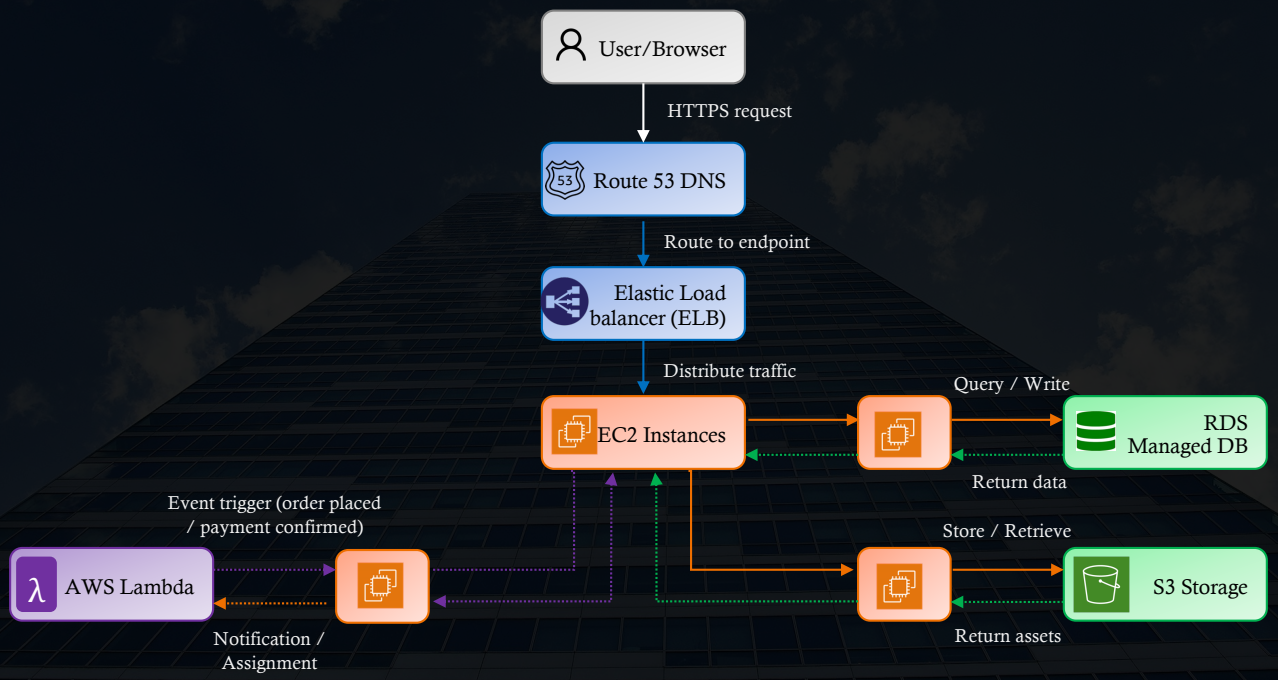
WHY AWS?

Provider	Ecosystem Fit	Load Balancing	Serverless	ABC Alignment
aws	Broadest managed-service catalogue	ELB – native Route53 integration	Lambda – event-driven, zero idle cost	Best fit – Route 53 already in stack
Azure	Microsoft / enterprise-aligned	Application gateway – extra config needed	Azure Functions – strong but separate ecosystem	No existing Azure signals in ABC
Google Cloud	Analytics and ML-first	Cloud load balancing – GKE-oriented	Cloud Run / Functions – container first	No analytics-heavy workloads in ABC now

AWS IN ACTION



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CONCLUSION

Public cloud | IaaS+PaaS | Hybrid cost model | AWS

Key Outcomes Unlocked:

- 80% cost reduction (measured service)
- 10x surge absorbed (rapid elasticity)
- Zero-headcount scaling (resource pooling + Auto Scaling)
- Workflow automation (Lambda event triggers)

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Thank you!

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